

Using Google Reviews to Identify Healthcare Access Barriers in Germany: Insights Relevant to Refugees and Migrants

Hausarbeit

zur Erlangung des akademischen Grades „Master of Science (M.Sc.)“ im Studiengang Wirtschaftsingenieur der Fakultät für Elektrotechnik und Informatik, Fakultät für Maschinenbau und der Wirtschaftswissenschaftlichen Fakultät der Leibniz Universität Hannover

vorgelegt von

Name:	Borchers	Vorname:	Finn
Geb. am:	07.09.2000	in:	Langenhagen
Matrikelnummer:	10035703	Fachsemester:	6
Adresse:	Nikolaistraße 5, 30159 Hannover		

Prüfer: Prof. Dr. Michael H. Breitner

Betreuerin: M.Sc. Fenja Schulte

Hannover, den 30.06.2026

Abstract

Existing healthcare data often fail to capture lived, publicly articulated, and experience-based barriers to care. This paper examines whether Google Reviews of hospitals can serve as a complementary digital data source for identifying access, communication, and service-related barriers in the German healthcare system, with particular relevance for migrant-sensitive healthcare.

Keywords: ...

Table of Contents

List of Figures	i
List of Tables	i
List of Abbreviations	i
List of Appendixes	i
Research Summary	ii
1. Introduction	1
1.1 Motivation	1
1.2 Research Gap	1
1.3 Research Need	1
1.4 Research Question	1
1.5 Aim and Structure of this Thesis	2
2. Theoretical Background	3
2.1 Migration and Healthcare Access in Germany	3
2.2 Technical Background	3
3. Research Design	4
3.1 Data Source	4
3.2 Data Collection	4
3.3 Topic Modeling and Taxonomy Development	7
3.4 LLM Based Analysis	8
3.5 Benchmark Design	10
3.6 Model Comparison	12
3.7 Batch Processing	14
3.8 Literature Review Strategy	15
3.9 Ethical Considerations	19
4. Results and Findings	20
4.1 Benchmark Results	20
4.2 Label Patterns	22
4.3 Metadata Patterns	24
4.4 Literature Results	25
5. Discussion and Implications	25
5.1 Methodological Implications	25

5.2 Substantive Implications	26
5.3 Practical Implications	27
6. Limitations and Further Research	27
7. Conclusion	28
8. References	31
9. Appendix	32
10. Declaration of Honor	33

List of Figures

Figure 1: Overview of AI-supported Mental Health Apps in Psychocardiology	2
Figure 2: Application of DSR to the development of LeA-AI .	Fehler! Textmarke nicht definiert.
Figure 3: Connected papers results graph for selecting Chiu (2024).....	16
Figure 4: Chat interface of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 5: Content/Modules interface of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 6: Module Details page of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 7: Video step page of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 8: Homework page of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 9: Homework Details page of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 10: Settings page of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 11: Dashboard of LeA-AI	Fehler! Textmarke nicht definiert.
Figure 12: Dashboard showing module structure of LeA-AI .	Fehler! Textmarke nicht definiert.
Figure 13: Dashboard showing module details of LeA-AI	Fehler! Textmarke nicht definiert.

List of Tables

Table 1: Overview of Literature	18
Table 2: "Overview of Chatbots, Applications, and Wearables", based on Pavlopoulos et al. (2024)	Fehler! Textmarke nicht definiert.

List of Abbreviations

“AI”: Artificial Intelligence

“ML”: Machine Learning

...

List of Appendixes

Appendix 1: Chat-Viewmodel in the Dart Programming Language.**Fehler! Textmarke nicht definiert.**

Appendix 2: LeA-AI Prompt.....**Fehler! Textmarke nicht definiert.**

Research Summary

The increasing global ...

The central question is as follows:

“What barriers to healthcare access for refugees and migrants in Germany are identified in the literature, and how can these barriers be structured into a coherent framework?”

The overarching aim of this thesis is to examine how ...

This study adopts a Design Science Research (DSR) approach, e...

Theoretical Background

Artificial Intelligence ...

Research Design

To address the c...

Literature Review

The current state of r...

Discussion & Implications

The integration o...

Limitations & Further Research

Despite

Conclusion

AI has ...

1. Introduction

1.1 Motivation

Deutschland als relevanter Kontext von Migration und Flucht (100 Mio. Menschen auf der Flucht weltweit, DE unter den Top 5 Zufluchtsländern, ...)

psychische Belastung und Zugangsprobleme (Höhe Prävalenz von Depressionen, PTBS, Anxiety), Es gibt beim Einreisen Tests für Infektiöse Krankheiten, aber nicht psychische Krankheiten...

Stigmatisierung, Unwissen bzgl. Versorgungslage (wo bekomme ich Hilfe, wie, was kostet das?) sprachliche Hürden, wenig Deutschkenntnisse

Online Informationen, Social Media als Informationsquelle -> potentiell interessant für das Identifizieren von Versorgungslücken, Hürden, Problemen, Herausforderungen

→ Social Media und Daten als Grundlage für das bessere Verständnis bzgl. Gesundheitsversorgung

1.2 Research Gap

Was ist die Lücke? Es gibt Paper bzgl. ... Social Media Nutzung, Auswertung von Reddit, X, andere Plattformen, Google Reviews in den USA

Datenlücken in Deutschland sind groß in Bezug auf Migration im Speziellen

Wenig erforscht: Spezielle Forschung um eine vulnerable Gruppe: Migranten in Deutschland, mithilfe von Internetdaten, die öffentlich zugänglich sind (Google Reviews)

Migrationsbezogene Daten häufig nicht systematisch erfasst

1.3 Research Need

Probleme werden nicht aufgezeigt, nicht systematisch genug, und deswegen....

Daten sammeln, systematisch, auswerten, ...

Versorgungslücken in der Gesundheitsversorgung in DE besser verstehen

1.4 Research Question

Building on the urgent public health need and the technological potential outlined above, this thesis seeks to investigate how structured data analysis of Google Reviews... can serve this need... .

Given the high prevalence of ... and mental health disorders, and the current lack of targeted, scalable solutions for this complex ... group, the central research aim is to explore the

To structure this investigation, the following guiding research question is proposed:

Primary Research Question (RQ1):

“How can Google Reviews of hospitals be used as a complementary data source to identify healthcare access barriers in Germany, particularly barriers relevant to migrant-sensitive care?”

Secondary Research Question (RQ2):

“How can AI be leveraged to gain deeper, more detailed insights in barriers across the health care system in Germany?”

This question aims to bridge the interdisciplinary gap between ... By doing so, this thesis will contribute to a deeper understanding of how intelligent technologies can be purposefully designed and responsibly integrated to address one of the most pressing and under-served challenges in contemporary healthcare.

1.5 Aim and Structure of this Thesis

This thesis investigates how ...

Erst Hintergrund erläutern, mit Problemen, Lücke, research question, literatur ergebnisse, dann unsere daten sammlung, aufbereitung, nutzung verschiedener methoden (LLMs, topic modelling, DL,) , Sterne bewertungen, Taxonomy (Schema mit dem man anzahl an likes/sternbewertung einfließen lässt, identifikation von kernherausforderungen/barrieren,

Diskussion auswerten

Limitationen (kleiner datensatz, könnte, müsste größer, weiter detaillierter, größere stichprobe, nur niedersachsen, nur DE, Bias, automatische Google Übersetzung und Lokalisierung der Bewertungen, Bias des LLM,), die Autoren (also wir) haben nicht alle relevanten paper gefunden, hätten noch mehr suchen können, andere keywords für andere suchergebnisse.....

further research... mehr daten, größer, weiter, besser, andere social media quellen, andere reviews, cross verglerich, anderer methoden,

Kurzer Überblick zur systematik hinter unserer auswertung

Figure 1: Overview of systematic data collection, preparation, analysis for healthcare system challenges

Given the overview of this topic, the structure of this thesis reflects this interdisciplinary and practice-oriented approach (Figure 1). Following the introduction, the theoretical

background chapter lays the foundation by covering both the technical aspects of AI, including

2. Theoretical Background

2.1 Migration and Healthcare Access in Germany

Seit 2015 erhöhte flüchtlingsbewegung, so viele flüchtlinge weltweit wie nie zuvor, DE unter den top ländern, die flüchtlinge aufnehmen,....

→ Keine direkte wiederholung der dinge aus der introduction

Social media/Internet nutzung von flüchtlingen, probleme von flüchtlingen
Internet als informationsquelle für flüchtlinge

→ Was kommt in die Haupt-Literatursuche rein? Also Kapitel 4??

→ Google Reviews als Datenquelle um health care barriers besser zu verstehen

(Brenner & Lok, 2022)

2.2 Technical Background

Advances in AI, LLMs, what is AI, why AI can help with better, deeper understanding and analysis of data,

AI in general

Machine Learning (ML)

Deep Learning (DL)

Large Language Models (LLM)

3. Research Design

3.1 Data Source

This study uses public Google Maps reviews of hospitals in Lower Saxony as its empirical basis. The platform provides an institution-linked review environment in which users publish short evaluative texts that are attached to specific healthcare providers. The resulting corpus therefore consists of publicly visible, location-specific, and institution-referenced review texts.

These reviews contain concise but information-rich descriptions of hospital experience. They refer to treatment processes, waiting times, communication with medical and administrative staff, perceived friendliness or disrespect, organizational coordination, and general impressions of care. Although the texts are often brief, they frequently condense concrete moments of satisfaction, frustration, praise, or conflict into linguistically compact narratives. For the present study, the review text constitutes the analytical unit, while Google Maps functions as the source environment in which these texts become publicly observable and institutionally attributable.

The data source is analytically relevant because it captures naturally occurring evaluative language that is not elicited through a questionnaire or interview guide. Review authors formulate their observations in their own words and emphasize those aspects of the hospital encounter that they regard as salient. This makes the corpus suitable for the identification of recurring positive and negative themes at scale, especially with regard to treatment, staff interaction, communication, waiting, and organizational processes.

Because the analytical value of this source depends on systematic and reproducible collection, the following section describes how the relevant hospital locations are identified, how the associated review pages are captured, and how the resulting review texts are prepared for structured analysis.

3.2 Data Collection

The empirical source defined in the previous section requires a dedicated collection procedure because Google Maps does not provide the complete review corpus through a static download interface. Official Google place resources expose structured place information, but only a small subset of reviews, and therefore do not permit corpus-complete retrieval for hospitals with large public review volumes. The study therefore uses an automated browser-based workflow that transforms publicly visible review content into structured research data.

The collection workflow is organized as a two-stage pipeline. In the first stage, hospital names are resolved into canonical Google Maps place pages. In the second stage, reviews are captured from those resolved place pages. This separation is

methodologically relevant because it distinguishes two different sources of uncertainty: place-resolution uncertainty and review-capture uncertainty. The former concerns whether a hospital name can be matched reliably to the correct Google Maps place. The latter concerns whether the reviews attached to that place can be loaded, parsed, and stored with sufficient completeness. Keeping these stages separate improves traceability and preserves a clearer audit trail.

The first stage begins with a plain-text list of hospitals and clinics. At the documented project state, `'configs/places.txt'` contained `'171'` clinic entries after excluding comments and blank lines. Each entry is converted into a stable internal identifier so that outputs can be stored consistently across repeated runs. The resolver opens a Google Maps search for each institution, waits for a stable place page, and extracts the resolved page title. To reduce the risk of accepting incorrect matches, the requested hospital name and the resolved page title are compared by means of normalization rules that account for capitalization, punctuation, spacing, and German character variants. The outcome of this step is written to a persistent lock file that records whether an entry is resolved, marked as a mismatch, or terminated with a technical error. In the documented extraction state, `'place_urls.json'` contained `'170'` place entries, of which `'126'` were resolved and `'44'` were marked as mismatch. These figures represent extraction-state indicators rather than substantive characteristics of the hospital landscape itself.

The second stage captures review content from the resolved place pages. This stage relies on the Chrome DevTools Protocol, which allows the script to control a live browser session and inspect a dynamically changing interface state while Google Maps loads content progressively. For each run, the script creates a temporary browser profile in order to reduce unintended effects from prior cookies, cache state, or session history. The browser language is set to German so that the interface state corresponds to the linguistic context of the corpus. After opening a resolved place page, the script handles consent interactions where necessary, identifies the control that opens the full review panel, and navigates the interface programmatically until the review container is accessible.

The identification of the review-opening control follows a heuristic strategy instead of relying on a single selector. Candidate controls are evaluated through a combination of semantic, structural, and behavioral cues, including review-related wording, visible review counts, JavaScript actions, and button-like interaction patterns. This design reduces dependence on volatile interface details and increases robustness against minor interface changes.

Because Google Maps reviews are loaded progressively, the script scrolls the review container iteratively and evaluates whether additional content continues to appear. Progress is monitored through several signals, including the number of unique review identifiers currently visible, changes in the height of the review container, the distance to the bottom of the panel, observed review-related network activity, and the declared

total review count where such a value can be inferred from the interface. The stopping logic follows a saturation-based termination procedure. The script stops when the declared total has been reached and the panel remains stable, when repeated bottom probes no longer generate growth, or when configured safety limits for scrolling or time are reached. Methodologically, it is necessary to distinguish between the declared review total shown by the interface, the loaded review total observed during scrolling, and the final number of saved review records after parsing and deduplication.

Once the review panel has reached a saturation state, collapsed text segments are expanded where possible and the resulting review view is stored as an intermediate HTML artifact for parsing. The parser extracts the final records from this saved review-panel HTML rather than from the limited official Places API. It then converts the visible entries into a standardized record structure. Each record stores the reviewer name, the review text, the star rating where available, and the relative review time shown in the interface. In addition, the parser captures visible engagement and response metadata, including the publicly displayed like count, the presence of an owner response, the displayed owner response time, and the owner response text where such information is available.

To increase robustness, the capture procedure supports repeated attempts for the same hospital page. Later attempts may use slower scrolling behavior in order to improve loading completeness on more demanding pages. Records obtained across attempts are merged and deduplicated before the final output is written. Deduplication primarily relies on review identifiers and falls back to normalized reviewer and text combinations when necessary. Where duplicate entries contain complementary information, the more complete variant is retained.

The final output of the collection stage is a structured per-hospital review corpus stored as a JSON array per clinic in 'reviews.json' files, accompanied by procedural metadata such as retry traces and batch-level run summaries including 'capture_reviews_run_summary.json'. In the documented artifact state, the corpus contained '122' review files with '31,823' extracted reviews in total, a median of '200.5' reviews per institution, and a range from '2' to '1,070'. These values are reported as time-bound extraction-state indicators, because Google Maps content and interface behavior change over time. The collection stage thus creates a structured and auditable corpus that can support the later comparative multi-method analysis.

The collection procedure is also bounded by responsible data-handling principles. Only publicly visible review content is captured. Private accounts, non-public records, and direct contact data are excluded. Engagement and owner-response fields are extracted only from publicly displayed interface elements, and the resulting corpus is used for descriptive analysis of public narratives rather than for the identification of individual persons.

3.3 Topic Modeling and Taxonomy Development

After the review corpus has been collected and standardized, the study introduces an intermediate analytical step before structured coding is applied. This step uses topic modeling to explore recurring semantic patterns in the review texts and to identify candidate thematic domains for the subsequent taxonomy-based analysis. Within the research design, topic modeling functions as an exploratory bridge between raw textual material and later structured extraction.

An exploratory step is methodologically appropriate because Google reviews are unsolicited narratives rather than responses to a predefined questionnaire. Review authors decide themselves which aspects of a hospital encounter to mention, and these descriptions vary considerably in wording, emphasis, and level of detail. A purely deductive coding scheme would therefore risk overlooking recurrent experiential domains that emerge directly from the corpus. Topic modeling provides an inductive view of how review texts cluster semantically before a fixed category system is imposed.

The study uses BERTopic for this purpose. The implementation combines the multilingual sentence-transformer model ‘paraphrase-multilingual-MiniLM-L12-v2’ for semantic embeddings, UMAP for dimensionality reduction, HDBSCAN for density-based clustering, and class-based TF-IDF for topic representation. This methodological stack follows established components in neural topic modeling and is suitable for semantically heterogeneous short-text corpora because it groups documents by semantic proximity while also allowing a substantial outlier region for texts that do not belong to a stable cluster.

The topic model is not fitted only once on the full corpus. Instead, separate models are constructed for analytically defined subsets, namely negative reviews (‘1-2★’), positive reviews (‘4-5★’), liked reviews (at least ‘1’ like), and highly liked reviews (at least ‘3’ likes). This design allows thematic structure to be examined within different evaluative and engagement-based segments of the corpus. A single global model would be more strongly shaped by the largest and most generic review patterns, whereas separate subset models preserve greater sensitivity to segment-specific topic constellations.

The topic-modeling stage is refined iteratively. The tuned configuration uses a minimum topic size of ‘35’ and HDBSCAN minimum samples of ‘5’. At the time of the documented run, the topic-modeling script identified ‘30,863’ reviews with non-empty text, including ‘11,943’ negative reviews, ‘17,700’ positive reviews, ‘18,055’ liked reviews, and ‘9,773’ highly liked reviews. Like-count metadata further indicate a mean of ‘2.731’ likes, a maximum of ‘172’ likes, and ‘9,146’ detected owner responses across the text-bearing review set. Outlier shares remained substantial across the subset models, ranging from ‘38.1%’ in the highly liked subset to approximately half of the documents in the other subsets. For heterogeneous short-text corpora, such outlier proportions are methodologically plausible and indicate that the exploratory model

captures only part of the corpus structure in the form of coherent topic clusters. The comparatively lower outlier share in the highly liked subset also indicates somewhat tighter semantic clustering in this segment.

The topic-modeling workflow produces three main output types: topic tables for each subset, review-level assignment tables, and summary metadata on corpus size, settings, and topic counts. In the current tuned run, these outputs are materialized as '`<subset>_topics.csv`', '`<subset>_assignments.csv`', and '`topic_model_summary.json`' in '`analysis/topics_tuned`'. These outputs are not treated as a final analytical result in themselves. Instead, they serve as an exploratory basis for taxonomy development.

The output of BERTopic is not used as the final taxonomy directly. The topic-modeling results are consolidated through an interpretive synthesis that proceeds by inspecting the largest and clearest topics, reading top terms together with representative review snippets, and merging semantically related topics into broader analytically meaningful categories. This consolidation step is necessary because topic labels frequently contain interpretable semantic signals while remaining linguistically noisy, fragmented, or locally phrased.

Conceptually relevant literature further sharpens this process. The topic model provides an inductive overview of the corpus, while literature helps stabilize the interpretation of access-related domains that may appear in different lexical forms or across several neighboring topics. The resulting category system is therefore data-informed and conceptually grounded. At this stage, literature is used to refine and delimit the taxonomy, not to displace the inductive structure of the corpus.

The output of this stage is a structured working taxonomy that contains both barriers and strengths and that is suitable for later operationalization in a large-language-model-based extraction workflow. The following section explains how this working taxonomy is translated into a structured LLM-based review analysis procedure.

3.4 LLM Based Analysis

Once the working taxonomy has been established, the next methodological step is to translate it into a structured review-level analysis procedure. The topic-modeling stage identifies recurring thematic domains, but it does not yet produce standardized annotations for individual reviews. The large language model stage addresses this gap by formulating the analysis of each review as a structured extraction task in which free-text narratives are mapped onto a predefined label space.

The analytical unit in this stage is the individual review. Each review may contain multiple positive and negative elements at the same time, which means that the task cannot be reduced to a single overall sentiment label. The model is therefore asked to identify which problem labels and which strength labels are present in the text. For the present corpus, simultaneous extraction of barriers and strengths is methodologically

preferable to single-label polarity classification because many reviews combine positive and negative assessments within the same narrative.

The distinction between taxonomy, schema, and output record is central at this stage. The taxonomy defines the conceptual label space. The schema defines how this label space is operationalized in a structured output format. The output record stores the extracted labels together with the review reference and metadata in a standardized analytical representation.

The initial implementation uses a nested mirrored schema. In this schema, ‘problems’ and ‘strengths’ are represented separately across the same domain families, including access, administration, communication, staff, care, coordination, environment, and inclusion. Across these mirrored domains, the schema comprised ‘34’ content fields per polarity. The purpose of this structure is analytical clarity, because praise and criticism can be located within the same conceptual domains while remaining structurally distinct. At the same time, the large number of nested fields creates additional formal complexity for the extraction step.

The extraction design is later simplified into a flat schema built around two canonical label lists, namely ‘problem_labels’ and ‘strength_labels’. This change does not reduce the conceptual scope of the taxonomy itself. It reduces the formal complexity of the structured output while preserving the same underlying label space. Methodologically, the flatter structure is better suited to consistent extraction, downstream comparison, and later aggregation across a large review corpus because the model no longer has to reproduce a large mirrored JSON hierarchy before assigning the substantive labels.

In its final operational form, this flat schema contains ‘34’ canonical labels distributed across ‘8’ domain families: access, administration, communication, staff, care, coordination, environment, and inclusion. The standardized output record stores ‘review_id’, ‘review_index’, the review text, associated metadata, and the two label lists ‘problem_labels’ and ‘strength_labels’. The final analytical representation is therefore conceptually rich at label level while remaining structurally compact at record level.

Prompt construction in this stage combines the review text with selected metadata. The review remains the central analytical object, but contextual metadata such as star rating, like count, and owner response information provide additional situational cues. These cues are intended to support more stable interpretation of evaluative context while keeping the extraction task anchored in the review text.

The model output is generated in structured form. The response is parsed into the expected extraction schema, validated locally against the predefined structure, and stored in a standardized review-level record format. Each record preserves the original review reference, the associated metadata, and the extracted problem and strength labels. This design ensures that later analytical steps operate on a consistent corpus representation.

Because the extraction output is standardized, different prompts, schemas, and model configurations can be compared against the same reference logic without changing the underlying analytical task. The LLM stage is therefore designed not only to produce review-level labels, but also to support benchmarking, comparison, and corpus-wide aggregation within a single methodological framework.

Once this structured extraction pipeline is defined, its quality must be evaluated against a reviewed reference set. The following section therefore explains how the benchmark dataset and its manual review workflow are constructed.

3.5 Benchmark Design

Once the structured extraction workflow has been specified, it requires a reviewed reference set for systematic evaluation. Without such a benchmark, prompt adjustments, schema decisions, and model selection would remain guided by unsystematic impressions. The benchmark is therefore introduced as a reviewed reference instrument for controlled comparison.

The benchmark consists of a stratified sample of ‘120’ reviews. The sample is balanced across three analytically relevant buckets, namely positive reviews, negative reviews, and mixed or difficult reviews, with ‘40’ reviews in each group. This design limits overrepresentation of one dominant review type and ensures that later evaluation covers both clear and ambiguous cases. The sampling notes further indicate that the benchmark spans multiple hospitals and limits overconcentration from any single institution.

The benchmark does not begin from an empty annotation layer. Instead, an initial set of model-generated labels is used as a starting proposal. These ‘model_prelabels’ are copied into a separate editable ‘benchmark_labels’ layer so that the reviewer can retain, remove, or add labels while preserving the distinction between model suggestion and reviewed reference annotation.

The manual review process is conducted in a dedicated local benchmark reviewer interface. This interface combines progress tracking, filtering, navigation, review text, contextual metadata, visible model suggestions, and editable benchmark labels within one workspace. The reviewer can therefore inspect each review together with its bucket assignment, star rating, like count, and owner response status while comparing the proposed labels with the review content, which supports traceable text-anchored revision of the benchmark set (Figure 1).

The first complete manual review of the ‘120’ benchmark cases took approximately four hours. The workflow proceeded from positive reviews to negative reviews and then to mixed or difficult reviews. This sequence corresponds to increasing interpretive ambiguity, because positive reviews are usually the least revision-intensive, negative

reviews require more problem-side differentiation, and mixed reviews require the most explicit balancing of positive and negative signals within the same text.

This reviewed benchmark is constructed through a single primary human review workflow that is supplemented by AI-assisted second review. It is therefore a curated reviewed reference set, not an inter-rater gold standard derived from multiple independent human annotators.

The workflow was later extended by an additional AI-based review assistance layer. This second review component does not change benchmark labels automatically. Instead, it provides a structured second opinion on the currently assigned labels by surfacing possible additions, possible removals, and short text-based reasoning anchored in the review content. The expanded interface displays suggested changes, critical text passages, and explanatory notes alongside the existing benchmark labels, thereby preserving human control over the reviewed reference set (Figure 2). In the final reviewed snapshot, all '120' benchmark cases contained an 'ai_review' block; '43' were in 'ready' status and '77' in 'stale' status, which indicates that the second-review layer remained embedded in an active manual correction workflow rather than replacing it.

The output of this stage is a stable reviewed benchmark snapshot derived from a benchmark base file and a final reviewed version. Each review receives a persistent review status, displayed in the interface as 'reviewed' and stored in the underlying JSON state as 'reviewed_by_user', and the first complete reviewed snapshot records '120/120' cases as reviewed. The documentation density of this snapshot is also substantial: '71' reviews contain explicit 'benchmark_notes', of which '34' refer directly to AI recommendations and '37' provide independent manual justification. The reviewed snapshot thus functions as a fixed evaluation reference for later prompt, schema, and model comparisons.

![[Figure 1](/Users/finnborchers/human-ls/human-ls/analysis/benchmark-reviewer/Bildschirmfoto 2026-05-14 um 09.23.32.png)]

Figure 1. Benchmark reviewer interface showing the current review, contextual metadata, model prelabels, and editable benchmark labels in one local review workspace. Source: own representation.

![[Figure 2](/Users/finnborchers/human-ls/human-ls/analysis/benchmark-reviewer/Bildschirmfoto 2026-05-14 um 11.59.26.png)]

Figure 2. Benchmark reviewer interface with expanded AI-based second-review pane, including suggested additions, suggested removals, and text-anchored reasoning. Source: own representation.

Once this reviewed reference set exists, prompts, schemas, and model variants can be compared against a common evaluation logic. The following section therefore explains how the subsequent model-comparison experiments are structured.

3.6 Model Comparison

Once the reviewed benchmark has been established, it becomes possible to compare alternative extraction setups systematically. The study uses this benchmark to evaluate prompt variants, schema choices, and model configurations under a common reference logic. The purpose of this comparison stage is to determine how individual methodological choices affect extraction quality, generalization, and operational robustness.

The original ‘V1’ configuration functions as the main methodological baseline for the subsequent experiments. In conceptual terms, this baseline combines a flat extraction structure, a fixed taxonomy of problem and strength labels, and a prompt that includes the review text together with contextual metadata. Later variants are designed as controlled deviations from this setup so that individual changes can be interpreted more clearly.

The experiment program does not begin directly on the full benchmark. Its first calibration stage is based on a fixed ‘12’-review sample that was constructed to cover positive, negative, and mixed cases, including communication problems, privacy issues, discrimination or racism, staff behavior, treatment quality, waiting, and food or ward-related conditions. Within this calibration setting, the initial nested-schema comparison tested ‘gpt-4.1-nano’, ‘gpt-4.1-mini’, and ‘gpt-5-nano’. This first-stage comparison established that ‘gpt-4.1-nano’ was too weak and inconsistent for the extraction task, while ‘gpt-5-nano’ remained overly conservative and produced partially empty extractions in the nested structure. ‘gpt-4.1-mini’ therefore emerged as the earliest robust candidate.

The second experiment block concerns prompt variants and later snapshot and few-shot controls. On the full reviewed benchmark, the prompt-variation line tests whether extraction quality changes when metadata are removed or reduced, or when the prompt is made more restrictive or more explicit. A later control line fixes the model snapshot and temperature in order to improve run-to-run comparability, and an additional line introduces few-shot examples in order to test whether explicit demonstrations improve label assignment. These controls help distinguish effects of prompt wording from effects of model context.

The third experiment block concerns fine-tuning and augmentation. Fine-tuning is introduced as a data-centered alternative to prompt variation, while paraphrase augmentation is used to test whether synthetic wording variation improves generalization. In the augmentation line, ‘40’ difficult training reviews are expanded by ‘2’ accepted paraphrases each, yielding ‘80’ synthetic examples. At this point, it becomes necessary to distinguish between in-sample benchmark fit and out-of-sample holdout generalization. The full-120 benchmark provides a diagnostic surface for in-

sample behavior, whereas the holdout20 subset provides the more relevant signal for generalization beyond the reviewed training split.

The holdout20 subset is kept separate from the reviewed training split and is used only for generalization checks. Its methodological role is to distinguish benchmark fit on seen reviewed cases from performance on previously unseen reviews. Once fine-tuning and augmentation are introduced, this separation becomes necessary because in-sample agreement alone is no longer an adequate indicator of broader extraction quality.

The fourth experiment block consists of holdout and base-model robustness checks. Additional runs with stronger or alternative base models test whether the main bottleneck lies in prompt design, schema structure, or the underlying model family. This broadens the comparison beyond one single provider or one single model generation. Throughout this stage, model choice is evaluated not only in terms of extraction quality, but also with regard to schema complexity and operational practicability. Early synchronous runtime projections and later batch-price comparisons are therefore treated as relevant design signals, even though quality remains the decisive criterion.

The fifth experiment block concerns structured-output and batch controls. After prompt-centered and data-centered variants have been explored, later control runs isolate infrastructure-level variables. In particular, the study tests whether explicit structured response formatting changes extraction behavior and whether batch execution introduces additional drift. This distinction is methodologically relevant because it separates changes in analytical quality from changes in the operational execution pathway.

The model comparison is interpreted through several linked criteria, including agreement with the reviewed benchmark, robustness on mixed and ambiguous reviews, generalization to unseen cases, and operational practicability for larger-scale application. Across the newer models and variants, the dominant failure pattern takes the form of overlabeling combined with label substitution into semantically adjacent categories. The purpose of the comparison stage is therefore not to optimize a single metric in isolation, but to identify an extraction setup that remains analytically defensible across several dimensions at once.

Selected benchmark plots summarize the progression from the original baseline to the later synchronous and batch control runs. These figures visualize the corresponding benchmark agreement patterns across the principal comparison points (Figures 3 to 6).

The comparison program establishes the methodological basis for selecting the later operational analysis path. The next section explains how this path is scaled to the complete corpus through batch processing.

3.7 Batch Processing

After the model-comparison stage has identified an operational extraction setup, this setup must be applied to the complete review corpus. This full-corpus step differs from benchmark evaluation because its purpose is no longer comparative testing on a limited reference set, but the large-scale production of review-level labels across all available records. Batch processing therefore constitutes the operational execution layer of the selected analytical workflow.

The full corpus is not processed as one single batch submission. Earlier large submission attempts exposed a queue constraint in the OpenAI Batch API and produced ‘token_limit_exceeded’ errors when several large jobs were enqueued in parallel. The binding operational limit was therefore the number of enqueued tokens rather than the nominal credit balance. The final batch design follows directly from this observed constraint.

The resulting full-run corpus contains ‘30,863’ reviews and is divided into two processing scopes, namely ‘10000_ids’ and ‘20863_ids’. Each scope is then subdivided into smaller batch files. The first scope is partitioned into ‘5 x 2000’ reviews, while the second scope is partitioned into ‘10 x 2000’ reviews plus one final batch of ‘863’ reviews. This staged partitioning keeps the submission units within an operationally manageable size range.

Execution proceeds through a serial autopilot workflow. For each scope, the system uploads one batch input file at a time, submits one batch job at a time, polls the job status every ‘300’ seconds, fetches and materializes completed outputs, and submits the next batch only after the previous one has settled. The artifact structure distinguishes between input batch files, job-state files, normalized outputs, and merged scope summaries. This organization preserves a transparent operational trail from submission through materialization to corpus-level aggregation.

Completed batch outputs are merged first at the scope level and then across the full corpus. This process yields stable merged artifacts for the labeled corpus and its summary metadata, including ‘all_scopes_labeled.json’ and ‘all_scopes_summary.json’. After materialization and merging, the final full run contains the complete set of ‘30,863’ labeled reviews with ‘0’ parse errors.

The operational footprint of this stage is documented explicitly. The observed API-side runtime is ‘44m 23s’ for ‘10000_ids’ and ‘3h 09m 42s’ for ‘20863_ids’, which results in a combined batch-processing time of ‘3h 54m 05s’. Across both scopes, the total token volume amounts to ‘27,230,780’ tokens, and the observed or reconstructed total cost is approximately ‘5.91 USD’. These timings refer to API-side processing windows, whereas the total operational runtime can be slightly longer because polling and materialization occur between cycles.

Under these conditions, the full-run output constitutes a coverage-oriented inference layer under benchmark-documented quality constraints. It enables corpus-complete descriptive analysis, but it does not erase the quality distinction between the benchmark-validated baseline and the large-scale operational path.

With the operational processing path now defined, the methodological framework can be completed by clarifying how relevant literature informs and stabilizes the analytical category system. The next subsection therefore turns to the role of literature within the overall research design.

3.8 Literature Review Strategy

The objective of this literature review is to develop a comprehensive understanding of how AI is currently applied in digital mental health applications, particularly those addressing depression, with the goal of identifying technologies and strategies that can be adapted to the context of psychocardiology. Given the limited number of existing applications or studies specifically focusing on psychocardiology, the search strategy is deliberately broad and centers on AI use in depression management more generally. This allows the identification of therapeutic approaches, interaction models, and system architectures that can inform the design and refinement of the LeA-AI artifact.

This literature review adopts a structured and transparent methodology to ensure methodological rigor, traceability, and relevance (Vom Brocke et al., 2020). Recognizing the iterative and exploratory nature of literature searches in complex, multidisciplinary domains such as AI in digital mental health, the review process is designed to align with what is described as an iterative review approach. This includes multiple cycles of searching, reading, refining search terms, and reapplying filters, a process particularly suited to fast-evolving fields like digital health. The review strategy also follows guidance on defining a clear search scope, using multiple complementary databases, and applying consistent inclusion and exclusion criteria to ensure quality and topical alignment (Vom Brocke et al., 2020).

To gather relevant literature, a structured and iterative search process is conducted using four academic tools: Consensus.app, Connected Papers, Google Scholar, and German National Library of Science and Technology (German: “Technische Informationsbibliothek”) (TIB). Each platform plays a distinct role in the review. Consensus.app is used as a deep semantic search tool, leveraging AI to retrieve highly relevant literature based on conceptual matching. It is especially helpful for identifying focused, peer-reviewed studies on specific AI use cases. Google Scholar provides a broader coverage of academic literature and helps discover alternative perspectives and recent publications that may not appear in domain-specific databases. Connected Papers is employed to visualize citation networks and explore related works in AI-supported mental health interventions (Figure 3).

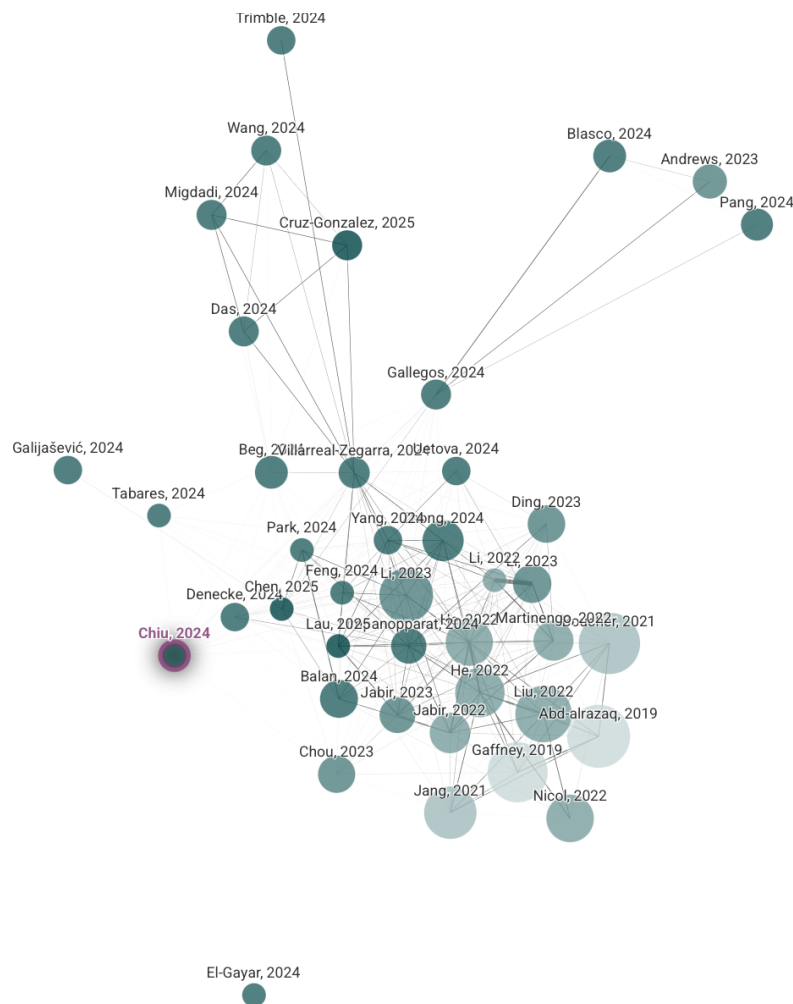


Figure 2: Connected papers results graph for selecting Chiu (2024)

Filters are applied across all platforms to ensure that only literature published within the past seven years is included, corresponding to the range of 2018 to 2025. This restriction is essential given the rapid evolution of AI technologies, particularly in health and human interaction contexts. The extended timeframe is deliberately chosen to capture the full arc of a transformative period in AI development. Beginning with the publication of “Attention Is All You Need” in 2017, which introduced the transformer architecture foundational to LLMs, and continuing through the years that saw their widespread application and societal impact (Vaswani et al., 2017). By 2020, the emergence of powerful LLMs like GPT-2 and GPT-3 significantly accelerated innovation in AI-supported communication, including applications in mental health. This timeframe also spans the COVID-19 pandemic, a period that not only heightened global awareness of mental health needs but also catalyzed the adoption of digital health technologies. Including both pre- and post-pandemic literature ensures that the review accounts for shifts in clinical practice, patient behavior, and the digital health ecosystem more broadly. Additional filters prioritize papers containing search terms within their title or abstract to ensure topical relevance.

Search strings are developed iteratively and include phrases such as “Refugees Germany”

(“AI in depression apps”,
“Artificial Intelligence in Psychiatry” and
“Artificial Intelligence depression treatment”).)

Boolean operators and variations are applied to refine the results further, and in the case of Consensus.app, where each query returns only a limited number of articles, the process involves repeated searches to expand the pool of literature. This ensures that even indirectly related but valuable papers are not overlooked.

The initial search yields a total of $n=120$ records (41 from Connected papers, 64 from Consensus.app, 6 from Google Scholar, 9 TIB). A multi-stage screening process is then undertaken. First, 11 duplicate records are identified and removed. Among the remaining 109 papers, 13 are excluded due to inaccessibility caused by paywalls or unavailable files. From the 96 accessible records, 5 are removed after abstract screening for lacking relevance to either AI or mental health applications. One paper is excluded for being in Spanish, as only English-language publications are considered. A final content-level review leads to the exclusion of 27 more papers that either lack empirical data or do not offer substantial contributions to the research question. This results in 64 papers being retained for in-depth analysis.

To address limitations in automated database algorithms, a set of $n=5$ key publications is added manually to the final set. These include widely cited foundational works in AI research and more recent empirical studies, which help contextualize the current state of AI in digital mental health applications in psychocardiology (Larionov et al., 2023; Vaswani et al., 2017). This manual inclusion strategy follows the precedent set by researchers, who emphasize the necessity of supplementing search results to compensate for the known gaps in algorithmic filtering (Tornero-Costa et al., 2023).

The 69 selected papers are then reviewed and categorized based on their primary application of AI. The categorization follows a concept-centric approach which emphasizes organizing literature by key themes or concepts rather than by author or chronology. This method supports the synthesis of knowledge by uncovering patterns, gaps, and relationships between conceptual areas. Here, it enables a clear comparison of how AI is applied across diagnostic, therapeutic, and operational domains in mental health interventions. By clustering the literature in this way, the review not only provides an organized summary of the current research landscape but also helps highlight underexplored intersections, such as the application of AI in comorbid cardiac and mental health settings (Webster & Watson, 2002). While many studies focus on diagnostic use cases, others examine treatment delivery and broader clinical support systems. As a result, three main thematic areas emerge: AI for diagnosis (e.g., speech- or image-based depression detection), AI for treatment (e.g., chatbot-based therapy or automated CBT modules), and AI for other use cases (e.g., through documentation, predictive analytics, or workflow assistance) (Table 1). It becomes clear that the majority of existing research concentrates on CBT-based interventions or on diagnostic tools. Very few studies focus on voice-based AI interactions for therapeutic purposes,

and none explore S2S communication or behavior planning within the framework of FACT.

Author	Year	Google Reviews	AI tools for analysis	Other data sources
Ali Husnain et al.	2024		x	
Beg et al.	2024		x	
Byeon et al.	2023	x		
Chatterjee et al.	2023	x		
Chen et al.	2023			x
Chiu et al.	2024		x	
Choudhary et al.	2024			x
Dhaker et al.	2023			x
Eid et al.	2023	x		
Elyoseph et al.	2024	x		
Gamble et al.	2020			x
Jan et al.	2018	x		
Larionov et al.	2023		x	
Li et al.	2023		x	
Mamidisetti and Reddy	2023	x		
Matías-Cristóbal et al.	2023			x
Mehta et al.	2021		x	
Nazir et al.	2022			x
Nelson et al.	2025		x	
Nyabuto and Nambiro	2024	x		
Pavlopoulos et al.	2024		x	
S. et al.	2023			x
Sadeh-Sharvit et al.	2023	x		
Schulte et al.	2024		x	
Souza Filho et al.	2021	x		
Su et al.	2020	x		
Suman et al.	2020			x
Turchioe et al.	2024		x	
Zaghloul et al.	2024			x

Table 1: Overview of Literature

The findings of the literature review directly inform the design decisions of the LeA-AI app. First, the strong emphasis in the literature on the challenges of user engagement and high dropout rates in digital mental health interventions supports the project's move toward a speech-first interface. Voice interaction significantly reduces usage friction and provides a more natural and accessible alternative to text-based systems. Second, the lack of existing studies on behavior experiments or AI-driven behavioral activation underlines the novelty of the LeA-AI approach, which integrates real-time planning of behavior experiments tailored to cardiac patients with comorbid mental illness. While previous literature confirms the efficacy of CBT in both face-to-face and digital formats, there is currently little research applying FACT specifically within AI-

supported environments. This underscores a significant theoretical and clinical gap that the LeA-AI artifact is uniquely positioned to address.

Overall, the literature review is conducted as a methodologically sound and iterative process, beginning with a broad search, followed by rigorous filtering, and finalized through thematic synthesis. From the initial 120 articles retrieved, 69 papers are ultimately included in the analysis. These provide a structured knowledge base that supports artifact development and highlights both well-established approaches and key areas where current research remains lacking. This foundation ensures that the design of the LeA-AI app is not only grounded in the most recent scientific understanding but also contributes to advancing the field through its focus on S2S interaction and behaviorally oriented therapy in psychocardiology.

3.9 Ethical Considerations

Given the sensitive google review data... namen, die haben wir ignoriert, personenbezogene daten wurden gefiltert...

4. Results and Findings

4.1 Benchmark Results

The benchmark compares model-predicted ‘problem_labels’ and ‘strength_labels’ against the reviewed benchmark labels. Within this evaluation logic, ‘exact matches’ denote review-level full agreement between predicted and reviewed label sets, ‘matched labels’ denote correctly predicted individual labels, ‘missing labels’ denote reviewed benchmark labels that are not predicted, and ‘extra labels’ denote predicted labels that are not present in the reviewed benchmark. The comparison sequence proceeds from the original baseline ‘V1’, through prompt variants ‘V1.1’ to ‘V1.6’, to fine-tuning and augmentation variants ‘V1.7’ to ‘V1.8’, and finally to the structured-output and batch controls ‘V1.9’ and ‘V1.10’.

The benchmark comparisons yield a consistent result pattern. The original ‘V1’ configuration remains the strongest observed baseline, and none of the later prompt variants surpasses it in general benchmark alignment. Among the later controlled prompt follow-up runs, ‘V1.5’ performs best, but still remains below the original ‘V1’ result. The intermediate variants ‘V1.1’ to ‘V1.4’ and the few-shot variant ‘V1.6’ all underperform relative to ‘V1’, despite differing tradeoffs between recall loss and overlabeling. Metadata removal, metadata reduction, and more restrictive prompt guidance do not produce an improvement in extraction quality.

The prompt-variation line therefore indicates that additional prompt structure does not improve calibration under the present task conditions. The observed benchmark pattern is more consistent with a task that is sensitive to prompt and model context changes than with a task that benefits from increasingly explicit prompt instructions.

The fine-tuning line yields a different pattern. ‘V1.7’ produces the strongest full-120 in-sample exact-match count with ‘53’ exact matches, which indicates that the fine-tuned model fits the reviewed benchmark cases more closely after training. However, the holdout20 comparison yields a sharply weaker result. On the unseen holdout reviews, ‘V1’ yields ‘8’ exact matches, ‘92’ matched labels, ‘18’ missing labels, and ‘5’ extra labels, whereas ‘V1.7’ yields ‘1’ exact match, ‘77’ matched labels, ‘33’ missing labels, and ‘25’ extra labels. ‘V1.8’ paraphrase-augmented fine-tuning improves slightly over ‘V1.7’, but remains clearly below ‘V1’. The augmentation setup expanded ‘40’ difficult training reviews by ‘80’ synthetic paraphrases, all of which passed the semantic judge (‘80/80’). Under the present setup, this process did not translate into convincing holdout gains, which indicates that the augmentation judge was more permissive than selectively corrective. The observed difference between full-120 fit and holdout20 performance is consistent with overfitting under the current training setup.

The holdout base-model sweep does not reverse this result pattern. Stronger or more expensive base models do not solve the task reliably on unseen reviews. Across the

holdout20 runs, the newer base models show weak exact-match counts and remain prone to overlabeling or unstable behavior on mixed and ambiguous cases. Within the present benchmark, the bottleneck is therefore not adequately explained by raw base-model capability alone.

The structured-output and batch controls isolate the most consequential technical regressions. ‘V1.9’ holds the ‘V1.5’ setup largely constant while introducing explicit ‘response_format = json_schema’. The benchmark result shows that matched-label counts remain almost unchanged while extra labels increase from ‘120’ to ‘174’. This pattern indicates that the explicit structured-output path is associated primarily with weaker label calibration rather than with a strong recall shift. Recall remains largely stable, whereas the dominant regression appears in the form of surplus labels. ‘V1.10’ then isolates the Batch API itself by holding the ‘V1.9’ conditions fixed and moving only to batch execution. The result is only slightly weaker than ‘V1.9’, which indicates that batching contributes, at most, a secondary effect relative to the explicit structured-output pathway.

Table 1 condenses the principal benchmark checkpoints across the original baseline, the strongest later synchronous control, the fine-tuned and augmented variants, and the later structured-output and batch controls.

Variant	Evaluation set	Exact matches	Matched labels	Missing labels	Extra labels	Interpretation
---	---	---	---	---	---	---
‘V1’	Full120	‘46’	‘577’	‘102’	‘44’	strongest overall baseline
‘V1.5’	Full120	‘26’	‘498’	‘181’	‘120’	strongest later synchronous control
‘V1.7’	Full120	‘53’	‘572’	‘107’	‘68’	strongest in-sample fit, but overfitting risk
‘V1.8’	Holdout20	‘3’	‘76’	‘34’	‘23’	slight improvement over ‘V1.7’, still weak on unseen data
‘V1.9’	Full120	‘19’	‘497’	‘182’	‘174’	regression concentrated in extra labels
‘V1.10’	Full120	‘18’	‘494’	‘185’	‘167’	small additional batch effect relative to ‘V1.9’

Taken together, the benchmark findings indicate that ‘V1’ remains the strongest observed baseline, that later prompt elaboration does not improve alignment, that fine-tuning currently improves in-sample fit more than out-of-sample generalization, and that explicit structured outputs constitute the main technical regression factor. The benchmark-review process further clarifies what human revision changed in the reference set itself. Across the ‘120’ reviewed benchmark cases, ‘76’ reviews underwent substantive label changes. Problem labels increased from ‘371’ to ‘431’, strength labels changed from ‘250’ to ‘248’, and the total label volume increased from ‘621’ to ‘679’. Revisions were weakest in the positive bucket, more strongly problem-oriented in the negative bucket, and most extensive in the mixed or difficult bucket. The most frequent additions included ‘care.competence’, ‘access.waiting’, ‘staff.seriousness’, ‘communication.information’, and ‘coordination.discharge’,

whereas frequent removals included ‘access.navigation’, ‘staff.empathy’, ‘staff.friendliness’, and ‘care.safety’. At domain level, the reviewed benchmark shifts relative weight toward communication, access, coordination, care, and inclusion. The selected benchmark plots summarize the benchmark agreement patterns across the principal comparison points from the original baseline to the later synchronous and batch controls (Figures 3 to 6).

![[Figure 3]](/Users/finnborchers/human-ls/human-ls/analysis/llm/benchmark_comparison/reviews_v1_120_plot.png)

Figure 3. Benchmark-agreement profile of the original ‘V1’ baseline. Source: own representation.

![[Figure 4]](/Users/finnborchers/human-ls/human-ls/analysis/llm/benchmark_comparison/reviews_v1_5_120_plot.png)

Figure 4. Benchmark-agreement profile of ‘V1.5’ as the strongest later synchronous control run. Source: own representation.

![[Figure 5]](/Users/finnborchers/human-ls/human-ls/analysis/llm/benchmark_comparison/reviews_v1_9_120_plot.png)

Figure 5. Benchmark-agreement profile of ‘V1.9’, isolating the effect of explicit structured outputs. Source: own representation.

![[Figure 6]](/Users/finnborchers/human-ls/human-ls/analysis/llm/benchmark_comparison/reviews_v1_10_120_plot.png)

Figure 6. Benchmark-agreement profile of ‘V1.10’, isolating the additional effect of batch execution under otherwise aligned conditions. Source: own representation.

4.2 Label Patterns

The full-run corpus contains ‘30,863’ labeled reviews and yields a differentiated overall composition. ‘1,005’ reviews (‘3.26%’) contain no labels, ‘9,528’ reviews (‘30.87%’) contain only problem labels, ‘13,347’ reviews (‘43.25%’) contain only strength labels, and ‘6,983’ reviews (‘22.63%’) contain both problem and strength labels. The mean number of labels per review is ‘3.59’, the median is ‘3’, and the maximum observed label count for a single review is ‘34’. Within the present corpus, this distribution indicates that neither purely positive nor purely negative reviews exhaust the data structure. The mixed-review segment remains large enough to be analytically consequential.

The histogram of labels per review documents the distribution of label density across the corpus, while the comparison of total problem and strength volume illustrates the asymmetry between negative and positive coding volume at corpus level (Figures 7 and 8). In retrospect, the mixed-review share is also consistent with the choice of a multi-label design, because a substantial segment of the corpus cannot be represented adequately through a single polarity assignment.

![[Figure 7]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/descriptive_analysis/plots/labels_per_review_histogram.png)

Figure 7. Distribution of total labels per review across the full labeled corpus. Source: own representation.

![[Figure 8]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/descriptive_analysis/plots/problem_vs_strength_volume.png)

Figure 8. Comparison of total problem-label and strength-label volume across the full corpus. Source: own representation.

The dominant thematic structure is concentrated around staff-related and care-related domains. On the problem side, the most frequent labels are 'care.treatment' ('5,013'), 'access.waiting' ('4,420'), 'staff.friendliness' ('4,088'), 'staff.respect' ('3,966'), and 'staff.seriousness' ('3,020'). On the strength side, the most frequent labels are 'staff.friendliness' ('15,723'), 'staff.empathy' ('9,742'), 'care.treatment' ('9,014'), 'care.competence' ('7,411'), and 'staff.respect' ('3,692'). The top-label plots document this concentration of recurrent themes around treatment, waiting, interpersonal conduct, empathy, competence, and respect (Figures 9 and 10).

![[Figure 9]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/descriptive_analysis/plots/top_problem_labels.png)

Figure 9. Most frequent problem labels in the full labeled corpus. Source: own representation.

![[Figure 10]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/descriptive_analysis/plots/top_strength_labels.png)

Figure 10. Most frequent strength labels in the full labeled corpus. Source: own representation.

The co-occurrence structure follows the same general pattern. The most common problem-problem pairs are 'staff.friendliness' plus 'staff.respect' ('1,678'), 'care.treatment' plus 'staff.respect' ('1,402'), and 'care.competence' plus 'care.treatment' ('1,359'). The most common problem-strength pairs are 'staff.friendliness' plus 'staff.friendliness' ('1,101'), 'access.waiting' plus 'staff.friendliness' ('1,088'), and 'care.treatment' plus 'staff.friendliness' ('929'). The co-occurrence heatmap documents that criticism and praise frequently cluster around the same experiential axes and that the same domains recur on both sides of the label space (Figure 11).

![[Figure 11]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/descriptive_analysis/plots/problem_label_cooccurrence_heatmap.png)

Figure 11. Problem-label co-occurrence structure in the full labeled corpus. Source: own representation.

4.3 Metadata Patterns

The most pronounced metadata signal in the labeled corpus is the star rating. ‘1★’ reviews are strongly problem-dominated, with an average of ‘3.18’ problem labels and ‘0.27’ strength labels, and ‘80.09%’ of them belong to the problem-only group. ‘5★’ reviews exhibit the inverse pattern, with an average of ‘0.21’ problem labels, ‘3.42’ strength labels, and ‘80.01%’ in the strength-only group. ‘3★’ reviews display the highest mixed-review share at ‘66.58%’. The star-distribution, review-mix, and average-label plots document this polarity structure and its relationship to label density (Figures 12 to 14).

![[Figure 12]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/metadata_analysis/plots/star_distribution.png)

Figure 12. Distribution of reviews across star-rating categories in the full corpus. Source: own representation.

![[Figure 13]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/metadata_analysis/plots/review_mix_by_star.png)

Figure 13. Distribution of problem-only, strength-only, mixed, and no-label reviews by star-rating category. Source: own representation.

![[Figure 14]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/metadata_analysis/plots/avg_labels_by_star.png)

Figure 14. Average problem-label and strength-label density by star-rating category. Source: own representation.

Like counts define a second descriptive pattern. Reviews with ‘0’ likes have an average star rating of ‘3.86’ and an average problem-label density of ‘1.03’. Reviews with ‘25+’ likes have an average star rating of ‘1.39’ and an average problem-label density of ‘4.51’. Intermediate like buckets follow the same direction: with increasing visible endorsement, average star ratings decline and problem-label density rises. Within the present corpus, like-count buckets are treated as descriptive visibility groupings rather than as indicators of factual validity or social consensus. The pattern is consistent with a platform selection mechanism in which more conflict-heavy reviews receive greater visible endorsement. The like-bucket plot documents this gradient directly (Figure 15).

![[Figure 15]](/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/metadata_analysis/plots/like_bucket_salience.png)

Figure 15. Average star rating and problem-label density across like-count buckets. Source: own representation.

Owner-response status yields a third descriptive grouping pattern. Reviews with owner responses have an average of '2.35' strength labels and a no-label share of '1.82%', whereas reviews without owner responses have an average of '1.99' strength labels, a no-label share of '3.86%', and an average like count of '3.13'. In the present analysis, owner-response status is treated as a descriptive grouping variable. The observed association does not support intervention claims about what owner responses do to review content and may likewise reflect platform-side selection mechanisms regarding which reviews receive a response. The owner-response profile plot summarizes this grouping pattern (Figure 16).

! [Figure 16] (/Users/finnborchers/human-ls/human-ls/analysis/llm/full_run/metadata_analysis/plots/owner_response_profile.png)

Figure 16. Descriptive profile of reviews with and without owner responses. Source: own representation.

Two additional boundaries of interpretation remain relevant. First, '56' reviews have no parsed star rating and are retained as an unknown category in order to preserve total-count consistency. Second, review time and clinic name are less suitable for primary next-step analysis because review times are mainly available as relative free-text strings and clinic names are not consistently preserved in the merged metadata. At the current analytical stage, the most robust metadata dimensions are therefore star ratings, like counts, and owner-response status.

4.4 Literature Results

Was andere gemacht haben, Google reviews, in DE, in den USA,
 Was haben die rausbekommen
 Was sind die hauptbarrieren in DE?
 Was findet man in google reviews, gibt es schon klinken analysen in der wissenschaftl. Literatur?

5. Discussion and Implications

5.1 Methodological Implications

The methodological contribution of the study lies in the combination of several analytical stages into one coherent design for public healthcare-review data. The design begins with corpus construction through browser-based collection, proceeds through exploratory thematic discovery by means of topic modeling, translates recurring themes into a structured taxonomy, operationalizes this taxonomy through review-level LLM extraction, validates extraction behavior against a reviewed benchmark, and finally applies the selected operational pathway to the full corpus.

This multi-stage structure is methodologically relevant because no single stage is sufficient on its own. Corpus construction without thematic exploration would leave the later label space under-specified. Topic modeling without structured extraction would remain exploratory. LLM extraction without a reviewed benchmark would leave model quality underdetermined. Benchmark validation without full-corpus inference would not provide corpus-level descriptive coverage. The design therefore depends on the interaction of corpus construction, exploratory discovery, structured extraction, benchmark-based validation, and full-corpus inference.

One of the main methodological implications concerns the distinction between validation quality and full-corpus coverage. The benchmark pathway identifies the strongest available extraction setup and makes quality differences visible. The full-corpus batch pathway applies a more operationally scalable configuration to the entire review population under explicit quality constraints. These two aims are related, but they are not identical. The study therefore supports a two-track methodological logic in which benchmark-grade validation and coverage-oriented corpus inference remain explicitly distinct.

5.2 Substantive Implications

The substantive findings indicate that public hospital reviews in this corpus are structured primarily by recurring themes of treatment, waiting, interpersonal conduct, communication, and coordination. These domains are not peripheral to the data. They form the main axes through which hospital encounters are publicly evaluated. This pattern is consistent with an understanding of patient experience that extends beyond treatment outcomes alone and includes process, interaction, and organizational interface conditions.

The mixed-review segment is substantively relevant because it captures ambivalence within patient and relative narratives. More than one fifth of the full corpus contains both problem and strength labels, and ‘3★’ reviews display the highest mixed share. This indicates that many reviews combine appreciation and criticism within the same account, often across the same experiential domain. For the substantive understanding of patient experience, these mixed narratives are informative because they preserve internal complexity that would disappear under a binary positive-versus-negative classification.

The observed like-count pattern adds a second substantive implication. Highly endorsed reviews are descriptively more negative and more problem-dense than the corpus average. Within the platform environment, visible endorsement therefore appears to operate as a mechanism of narrative salience rather than as a marker of factual validity. In substantive terms, this means that highly visible negative experiences occupy a disproportionate place in the public review ecology and may shape the collective visibility of hospital criticism.

5.3 Practical Implications

At the practical level, the resulting corpus and label structure support several downstream uses. First, they enable corpus-level monitoring views that summarize recurring strengths and barriers without relying exclusively on raw keyword search. Second, they support prioritization procedures in which reviews can be filtered by polarity, mixed character, or salience-related variables such as like counts. Third, the co-occurrence structure supports issue clustering by identifying which problem and strength dimensions repeatedly appear together.

These practical implications remain descriptive in scope. The present pipeline is suitable for corpus overview, review prioritization, and structured qualitative follow-up. It is not sufficient for institutional ranking or performance attribution. Its applied value lies in organizing large volumes of patient-facing narrative material into interpretable thematic structures that can support subsequent inspection and decision preparation.

6. Limitations and Further Research

The first set of limitations concerns the underlying data source. Google Maps reviews are self-selected public narratives and do not represent the full patient population of a hospital. The corpus therefore captures articulated public experience, not a representative distribution of all encounters. In addition, the reviews are subjective accounts rather than verified clinical records. Their analytical value lies in how experiences are publicly described, not in establishing factual truth about every reported event.

The second set of limitations concerns annotation and inference quality. The study identifies a clear difference between benchmark-validated quality and full-corpus batch coverage. The strongest benchmark baseline remains the original synchronous ‘V1’ configuration, whereas the full run depends on a more operationally scalable path with measurable quality loss. The full-corpus outputs therefore constitute structured descriptive evidence, but they should not be treated as benchmark-equivalent gold labels. In addition, the reviewed benchmark is based on one primary human review workflow supplemented by AI-assisted second review rather than on independent multi-rater adjudication. The reference set is therefore best interpreted as a carefully curated reviewed benchmark, not as an inter-rater gold standard.

The third set of limitations concerns metadata. Likes appear to function as a salience marker for visible negative narratives, but they do not measure objective importance or factual correctness. Owner responses describe a selection pattern in the corpus, but they do not justify causal claims about institutional responsiveness or intervention effects. Review times are available mainly as relative free-text expressions, which

restricts temporal analysis to coarse grouping. Clinic-level comparison is also limited because the merged outputs do not preserve a sufficiently stable clinic-level key for the present analytical layer, and any clinic ranking would risk overinterpretation under unequal review volume and variable capture completeness.

The fourth set of limitations concerns the boundaries of inference. The present workflow is designed for corpus characterization, thematic mapping, and methodological evaluation. It does not support clinic ranking or causal inference. Institutional performance measurement would require additional validation, cleaner metadata integration, and stronger safeguards against aggregation bias.

These limitations define several directions for further research. Future work should expand the reviewed training material, improve quality control for synthetic augmentation, and test whether better task-adapted examples reduce overlabeling and label substitution. A second line of work should strengthen large-scale inference quality while preserving operational efficiency. A third line should improve metadata integration, especially for clinic-level and time-sensitive analyses, provided that these dimensions can be captured and validated more consistently. Taken together, these steps would help reduce the gap between benchmark quality and full-corpus coverage.

7. Conclusion

This study develops a reproducible pipeline for converting public Google Maps hospital reviews into a structured corpus for the analysis of patient-facing hospital-experience narratives in Lower Saxony. The contribution begins with browser-based corpus construction, continues through exploratory topic modeling and taxonomy development, and culminates in review-level LLM extraction, benchmark validation, controlled model comparison, and large-scale batch execution.

Methodologically, the study demonstrates that public hospital reviews can be transformed into a transparent analytical dataset when URL resolution, review capture, stopping logic, parsing, benchmarking, and full-run execution are documented systematically. It also demonstrates that the strongest benchmark-quality extraction path is not identical to the most scalable corpus-wide execution path. The explicit separation between validation-oriented benchmarking and coverage-oriented corpus inference constitutes one of the central methodological contributions of the study.

Empirically, the labeled corpus documents a stable thematic structure centered on treatment, waiting, staff interaction, empathy, respect, competence, and coordination. Positive, negative, and mixed reviews all contribute to this structure, and highly liked negative reviews emerge as a salient subset within the public review environment. These findings do not support causal performance judgments about hospitals, but they

do make recurring access-related and care-related experience patterns visible at scale.

The study therefore contributes a structured approach for analyzing how hospital care is publicly experienced, evaluated, and made visible in a large digital review environment. In this sense, public Google Maps reviews do not replace conventional healthcare data, but they extend the methodological repertoire available for the descriptive analysis of patient-facing hospital-experience narratives.

8. References

- Brenner, S., & Lok, V. (2022). "We assist the health system doing the work that should be done by others" – a qualitative study on experiences of grassroots level organizations providing refugee health care during the 2015 migration event in Germany. *BMC Health Services Research*, 22(1), 309.
<https://doi.org/10.1186/s12913-022-07683-2>
- Larionov, K., Petrova, E., Demirbuga, N., Werth, O., Breitner, M. H., Gebhardt, P., Caldarone, F., Duncker, D., Westhoff-Bleck, M., Sensenhauser, A., Maxrath, N., Marschollek, M., Kahl, K. G., & Heitland, I. (2023). Improving mental well-being in psychocardiology—A feasibility trial for a non-blended web application as a brief metacognitive-based intervention in cardiovascular disease patients. *Frontiers in Psychiatry*, 14, 1138475. <https://doi.org/10.3389/fpsyt.2023.1138475>
- Tornero-Costa, R., Martinez-Millana, A., Azzopardi-Muscat, N., Lazeri, L., Traver, V., & Novillo-Ortiz, D. (2023). Methodological and Quality Flaws in the Use of Artificial Intelligence in Mental Health Research: Systematic Review. *JMIR Mental Health*, 10. <https://doi.org/10.2196/42045>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is All you Need. *Advances in Neural Information Processing Systems*, 30.
<https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>
- Vom Brocke, J., Hevner, A., & Maedche, A. (2020). Introduction to Design Science Research. In J. Vom Brocke, A. Hevner, & A. Maedche (Eds.), *Design Science Research. Cases* (pp. 1–13). Springer International Publishing.
https://doi.org/10.1007/978-3-030-46781-4_1
- Webster, J., & Watson, R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. *MIS Quarterly*, 26(2), xiii–xxiii.

9. Appendix

10. Declaration of Honor

I hereby certify that I have written this thesis independently and have not used any sources or aids other than those indicated, that all passages in the thesis which have been taken verbatim or in spirit from other sources are marked as such, and that the thesis has not yet been submitted in the same or similar form to any examination authority.

Hanover, 30.06.2026



Place, date*

Signature

*(date of submission)

Hiermit versichere ich, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe, dass alle Stellen der Arbeit, die wörtlich oder sinngemäß aus anderen Quellen übernommen wurden, als solche kenntlich gemacht sind und dass die Arbeit in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegt wurde.

Hannover, 30.06.2026



Ort, Datum*

Unterschrift

*(Abgabedatum)